# Coasters and pendulums

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use height differences to predict speed in ideal motion.

**SUCCESS CRITERIA - I can...**

1. Identify the relevant vertical height change.
2. Conserve energy only when the stated facts allow it.
3. Separate path length from gravitational energy change.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Gravitational potential energy changes from **-15 J** to **-3 J**. Find ΔU and state whether energy is gained or lost.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** mg Δh = ½mv²; v = √(**2 g** Δh).
Point mass released from rest, no losses or rotation; Δh is the positive vertical drop.

**Worked reasoning:** A pendulum released from rest **0.8 m** above its lowest point reaches speed √(**2** gh)=**4 m/s** there with g=**10** and negligible losses.

**Your turn:** Sketch the path and energy bars at the release, bottom and opposite turning point.

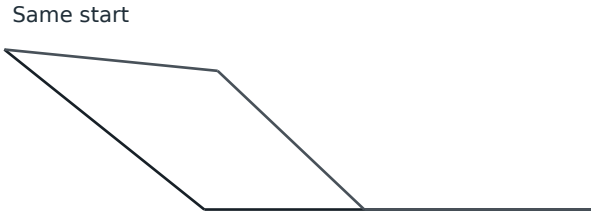
**Initial** **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

Two frictionless tracks start and end at the same heights but take different paths. **Compare final speeds** from rest.




Mark the vertical height change.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A **2 kg** cart drops **3 m**.
1. Use g=**10**.
2. **Find lost potential energy** and gained kinetic energy.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

**Initial** **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

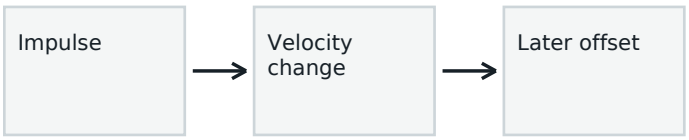
## 05 Explain your reasoning

A classmate uses total ramp length as h in mgh. **Explain how** to **name the correct measurement**.

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A fixed impulse changes a target's sideways velocity by **0.04 m/s**. **Find the predicted shift after 600 s** and **state a limitation**.
**Label the quantities** in the diagram and say what it treats as true.




**Marzano check - circle:** 1 2 3

## 07 Exit check

A pendulum loses **12 J** of gravitational energy while moving down. Neglect losses. What is its kinetic energy gain? **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

**Initial** **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

A **1 kg** pendulum bob is released from rest **0.8 m** above the bottom, but reaches only **0.6 m** on the opposite side. Use g = **10 N/kg**. **Complete the turning-point energy account** and **find the decrease** in mechanical energy.

**Initial** **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3